MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FIRST SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG1MDCSTA101 - Data Analysis using Libre Calc

Duration: 1Hour Maximum: 35 Marks

PRACTICAL EXAMINATION

From the following **eight** questions, students must answer any **five** questions. Each question carries **seven** marks.

Solve the questions using LibreOffice Calc in computer lab. The data in the question along with their answer must be written in the answer paper.

1. Generate 30 uniform random numbers in between 0 and 500. Find its mean, median and mode. Also find the range and standard deviation of the 30 randomly generated numbers.
   [Understand][CO2]

2. The weekly sales report of a shop in a village in Kottayam District is given below.

Sales Report

Sl.no	Item (unit)	No.of units	Unit cost
1	Pen (packet)	32	28.90
2	Pencil (packet)	57	32
3	Binder clip (box)	100	46.28
4	Floor cleaner (bottle)	70	94
5	Curry powder (packet)	100	46
6	Soap (packet)	135	50
7	A4 Paper (bundle)	20	150
8	Milk (packet)	550	28
9	Talcum powder (pack)	17	54
10	Shampoo (sachet)	64	3

Enter the above data in LibreOffice Calc and prepare one more column showing the total sales for each item (product of last two columns) and obtain the value of total weekly sale of the shop. [Apply][CO1]

3. The following data provides the number of different surgical operations performed in a hospital during a given week. Represent the data by means of a simple bar diagram

Type of operation	Number of cases
General surgery	110
Thoracic surgery	20
Prostate surgery	45
Urologic surgery	82
Surgery of abdomen	120
Neurosurgery	30
Others	18

[Analyse][CO2]

4. Thirty students in a business statistics class reported their major fields of studies, which is summarized as follows: Construct a simple bar diagram for the major fields for these 30 students.

Major field	Number of students
Accounting	12
Management	6
Marketing	4
Finance	5
Undecided	3

[Analyse][CO2]

5. The following data gives the time (in hours) taken for daily activities by a certain computer professional in a company. Represent the data by means of a simple bar diagram.

activity	work	sleep	leisure	reading	purchase	phone
time	9	6	3	1	1	4

[Analyse][CO2]

6. The following data provides the market sales of different laptop brands in a city. Represent the data by means of a pie diagram.

brand	A	B	C	D	E	F	G
sales	48	33	60	95	15	24	55

[Analyse][CO2]

7. The following data is about the monthly expenditure of two families A and B. Represent the data by means of two pie diagrams.

Item of expenditure	Expenditure(Rs.)	
	Family A	Family B
Food	210	230
Clothing	130	150
House Rent	100	130
Education	70	80
Medical	40	50
Others	50	80

[Analyse][CO2]

8. The following data shows the types of pets owned by different families in a city. Represent the data by means of a pie diagram.

Pet type	No. of families
Dogs	55
Cats	45
Love birds	50
Fish	30
Parrot	10
Others	10

[Analyse][CO2]

9. The following data typically represents the yearly expenditure of a student in a private university. Represent the data by means of a pie diagram.

Expenditure	Amount
Tuition	13600
Room rent	9000
Books Purchase	2400
Transportation	900
Recreation	2000
Other	2100

[Analyse][CO2]

10. The following data provides the number of tourists who visited India in December 2023 from 25 different countries.
50, 76, 30, 40, 78, 27, 39, 27, 67, 56, 52, 54, 27, 40, 86, 17, 97, 43, 34, 67, 86, 27, 39, 47, 40.
Calculate mean, median and mode for the above data. [Analyse][CO3]

11. The following data gives the daily wages (in rupees) of 30 workers in a factory:
60, 43, 40, 32,47,32,47,45,50,37,53,17,26,39,59,68,44,30,25,32, 40,62,46,29,32,54,41,32,30,39.
Calculate mean, median and mode for the above data. [Analyse][CO3]

12. The monthly salaries of 10 employees of a company are given below.

Sl.No	Name	Gender	Monthly Salary(Rs)
1	Varun Kumar	Male	41000
2	Chanchal Sawant	Female	25000
3	Rohan Patel	Male	32000
4	Rubeena Amir	Female	48000
5	Ann Thomas	Female	37000
6	Priya Mishra	Female	51000
7	Purushotham Sharma	Male	47500
8	Aashiq M	Male	43750
9	Megha Varma	Female	32000
10	Albert James	Male	32000

a) Find the mean, median and mode of the monthly income.
b) Find the mean and median of the monthly income of male and female separately. [Analyse][CO3]

13. An IT company manager tracks the working hours of 30 employees in a particular day as follows:
10, 11,7, 8, 10, 9, 8, 8, 7, 6, 10, 9, 8, 7, 10, 12, 9, 7, 6, 5, 9, 10, 11, 12, 11, 8, 5, 6, 10, 12 Find the range and standard deviation of working hours of the employees. [Analyse][CO3]

14. The time (in seconds) taken by 30 athletes to complete a 100 metre race are as follows:
15, 17, 22, 20, 14, 16, 15, 14, 20, 24, 16, 27, 18, 16, 24, 18, 12, 21, 24, 19, 15, 16, 15, 23, 28, 20, 18, 22, 24, 25. Find the range and standard deviation of the athlete's finishing time for that race. [Analyse][CO3]

15. The number of smart phones sold at 30 shops in Cochin on a particular day are as follows:
9, 16, 20, 14, 23, 13, 10, 22, 18, 9, 13, 8, 23, 22, 17, 15, 19, 12, 10, 9, 17, 19, 14, 13, 11, 23, 25, 21, 17, 15. Find range and standard deviation of phone sales occurred on that day [Analyse][CO3]

16. An IQ test was administered to 15 students before and after they were trained. The results are as follows. Use paired t test, examine whether there is any effect for the training programme.

Candidate	IQ score before training	IQ score after training
1	90	86
2	91	90
3	95	95
4	92	94
5	89	95
6	76	81
7	83	88
8	81	85
9	95	99
10	91	79
11	88	92
12	89	90
13	97	99
14	88	97
15	92	87

[Apply] [CO4]

17. A farm machinery manufacturing company has designed a new seed drill. In order to study the efficiency of the seed drill as compared to the existing one, an experiment was carried out with plots of land grouped in to 12 pairs. One of them was sown using special drill and the other by ordinary drill. The grain yields are given as follows: The company claims that their new seed drill increases the yield. Use paired t test, examine whether the data provide an evidence for or against their claim.

Plot pair no	Special drill	Ordinary drill
1	7.9	6.5
2	8.4	8.0
3	6.4	6.4
4	8.6	7.5
5	5.7	5.6
6	5.5	6.2
7	5.9	6.4
8	7.3	6.9
9	6.1	6.8
10	8.3	7.7
11	7.3	6.0
12	6.9	6.2

[Apply] [CO4]

18. The hours of spending in front of TV and test paper marks of 10 school students are as follows: Find Pearson's coefficient of correlation and interpret the result. Also represent the data by means of a scatter diagram.

TV hours	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5
TP marks	89	78	75	56	55	52	44	37	31	25

[Apply] [CO4]

19. The following data relates to prices of 10 commodities and quantities demanded. Find Pearson's coefficient of correlation between price and demand and interpret the result. Also represent the data by means of a scatter diagram.

Price	10	14	15	30	25	27	32	18	50	5
Demand	68	19	37	46	55	52	44	37	10	10

[Apply] [CO4]

20. The following data provides the mental age(x) and scores in a test (y) of a group of 10 students in a class. Calculate Karl Pearson correlation coefficient between ages (x) and scores (y) of students and interpret the result. Also represent the data by means of a scatter diagram.

age	10	12	13	14	16	17	18	5	7	9
score	24	20	23	24	18	19	20	1	8	20

[Apply][CO4]

21. The following data provides the length (x) in cm and weight (y) in gms of a sample of 10 articles in a factory. Calculate Karl Pearson correlation coefficient between length (x) and weight (y) of articles and interpret the result. Also represent the data by means of a scatter diagram.

length	4	12	10	20	16	9	10	9	12	3
weight	19	10	13	24	18	19	12	16	14	8

[Apply][CO4]

22. The following data relates to marital status and performance in an examination. Use chi-square test, examine whether the performance depends on marital status and interpret the result.

	performance	
	Good	Bad
married	70	90
unmarried	30	50

[Apply][CO4]

23. The following table shows the association among 1000 school boys of their general ability and their mathematical ability. Use chi-square test, examine whether general ability and mathematical ability are independent.

	General ability			
		Good	Fair	Poor
Mathematical ability	Good	57	20	3
	Fair	260	251	154
	Poor	59	94	102

[Apply][CO4]

24. The following data gives the yield of wheat in 30 test plots which are given 3 different fertilizers: Use ANOVA, test whether the fertilizers are equal in their effects.

Fertilizer1	80	70	65	60	60	50	75	80	75	85
Fertilizer2	80	75	70	65	60	60	70	75	80	85
Fertilizer3	60	60	60	50	50	40	65	75	70	70

[Apply][CO4]

25. The varieties A, B, C, D, E, F, G of paddy were sown in 5 plots each and the following yields in quintals per acre were obtained. Use ANOVA, test whether the data indicate a significant difference in the yields of the varieties.

A	B	C	D	E	F	G
13	10	12	15	13	13	14
12	12	11	10	10	11	12
18	16	13	17	14	15	19
15	13	14	14	15	16	17
11	11	10	10	9	12	15

[Apply][CO4]